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An optimized ensemble model for predicting average localization error of wireless sensor networks



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ABSTRACT

Wireless sensor networks (WSNs) are widely utilized in various applications due to their compact size, cost-effectiveness, and ease of deployment. Nonetheless, one of the biggest problems in WSNs is getting a reasonable estimate of the average location error of a node at the setup in the least amount of time. Wireless sensor networks can undergo changes over time due to various external and internal factors, such as environmental conditions, network congestion, hardware failures, or software updates. When these changes occur, the network may require redesigning, which can incur significant expenses. Traditional WSNs approaches, on the other hand, have been explicitly programmed, which makes it hard for networks to respond dynamically. Therefore, machine learning (ML) techniques can be used to respond appropriately in such scenarios. In this work, we proposed an optimized ML ensemble model for (i) identifying the critical network parameters for node localization when setting up wireless sensor networks with the accuracy needed in a short amount of time and (ii) predicting the average localization error of wireless sensor networks. We used the random forest algorithm with optimized hyperparameters from different optimization techniques to predict average localization error (ALE) using independent features like node density, anchor ratio, transmission range, and iterations.

1. Introduction

In recent years, wireless sensor networks (WSNs) have garnered significant interest due to their wide-ranging applications in environmental monitoring, industrial automation, agriculture, healthcare, coal mine [1–7] and significant advancements in the spread of ubiquitous sensing devices [8]. Sensor node localization (SNL) is a crucial component of WSNs but has also been acknowledged as one of the most challenging issues to address [9–16]. SNL is the process of figuring out where each node in the network is physically located. The objective of localization in WSNs is to determine the precise position of sensor nodes on a two-dimensional plane [17]. Localization in WSNs facilitates the provision of information to support decision-making based on the aggregated data transmitted from packets to base stations [17]. Also, the accuracy of node localization affects how well the network works, so reducing localization errors is very important. In the field of wireless sensor networks, predicting the average localization error (ALE) of WSNs is an important task that has been talked about. Mathematical models are one of the most common techniques used for node localization optimization [9,11]. These models consider things like the transmission power, the communication range, the presence of obstacles, and multipath fading, which are all things that affect the process of localization. On the other hand, this approach is thought to be hard to compute because the model needs to be tuned regularly as the input parameters change. To address this issue, current research [2,12,14,18,19] investigates how machine learning (ML) and quantum [20] can be used to better WSNs node localization optimization.

Tripathy and Khilar [21], proposed an ensemble strategy that merges Distance Vector-Hop (DV-Hop) and a weighted amorphous algorithm with the aim of enhancing localization precision or wireless sensor network. They compute dual distance measurements between unfamiliar and reference nodes, considering hop count and size. These measurements were subsequently fine-tuned through probabilistic distance estimation. The newly proposed method was subjected to a comparative analysis against conventional amorphous algorithms and three other improved amorphous techniques, showcasing its superior accuracy in metrics such as mean absolute error, mean squared error, and root mean square error [21]. Płaczek [22] presents a technique for minimizing data transfers in sensor networks by selecting certain nodes to carry out monitoring activities. Zhang et al. [23] suggested a method for localizing energy-efficient mobile nodes utilizing cooperative virtual array technology and a signal arrival and interaction algorithm. The study findings demonstrated a significant decrease in energy usage and strong performance in situations involving mobile nodes. Likewise, a Triangulation (DTN) algorithm that unites a WSNs with a genetic programming (GP) approach to enhance location algorithm accuracy was presented by Adama et al. [24]. Empirical findings validate GP's

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proficiency in predicting received signal strength index (RSSI) patterns. The proposed triangulation technique, in conjunction with GP, attains the lowest mean-distance error in real-time operations. This Triangulation method surpasses alternative localization algorithms, wherein anchor nodes gauge RSSI levels from neighboring nodes, and DTN employs RSSI data to estimate the proximity of neighboring nodes and form DT triangles. Mani et al. [25] proposed an iterative bounding box approach that was improved by including a Kalman filter to boost the accuracy of estimating the position of the unknown node in WSNs.

This study seeks to tackle the pressing challenges encountered in WSNs, especially promptly and precisely estimation of the average localization error during setup. Conventional methods have difficulties in adjusting to the ever-changing characteristics of WSNs, which requires a transition towards more flexible responses facilitated by machine learning techniques. Literature [21,26] explores the use of ensemble models to estimate the ALE of WSNs and suggests that effective hyperparameter tuning can optimize their performance. Nonetheless research shows that little or no work has been done experimentally to see how different optimization algorithms affect the ability of ensemble learners to make better node localization optimization. Therefore, this research focuses on predicting the average localization error for wireless sensor network nodes through an optimized ensemble model. It addresses the task of effectively pinpointing the best ensemble learner hyperparameters utilizing various ML optimization techniques to improve ALE prediction accuracy within a constrained timeframe. Specifically, we:

1. identify the influential network parameters for node localization when setting up wireless sensor networks with the accuracy needed in a short amount of time.
2. examining how different hyperparameters optimization algorithms affect the effectiveness and sensitivity of random forest algorithm.
3. propose a strong and well-tuned random forest algorithm that can predict the average location error of wireless sensor networks with high precision.

We presents an enhanced machine learning ensemble model that efficiently detects important network parameters during setup and accurately forecasts the average localization error. Our work utilizes advanced optimization approaches and feature-driven modeling to improve the dependability, efficiency, and flexibility of WSNs in real-world circumstances, providing new and valuable insights. This study offers several benefits for wireless sensor network localization. These include rapid identification of important parameters, accurate prediction of localization error, cost and time efficiency by reducing the need for expensive redesigns, robust feature-driven modeling, and improved reliability and efficiency in real-world situations. Our approach utilizes advanced machine learning techniques to quickly and accurately determine the location of nodes in wireless sensor networks (WSNs). This improves the overall performance and flexibility of the networks while keeping costs low. The remaining section of this paper is organized as follows: Section 2 presents a summary review of pertinent literature. In Section 3, we explain how the proposed strong and well-tuned random forest model for predicting the average location error of wireless sensor networks can be made to work. Section 4 presents our experimental results as well as related discussions. We end with conclusions and future work in Section 5.

2. Review of related works

In WSN, ALE is a measure of how well the localization process works. It is the average distance between where a sensor node is thought to be and where it really is. Therefore, predicting the ALE is an important task that has been the subject of much research in recent years in the field of WSNs [1,21,27,28]. Fig. 1 shows the categorization of node localization optimization approaches in WSNs [9]. For a more in-depth exploration of the various models for node localization optimization, we recommend referring to these works [9,13,29,30]. However, the following section of this paper discusses a few current studies utilizing these methods for node localization optimization accuracy.

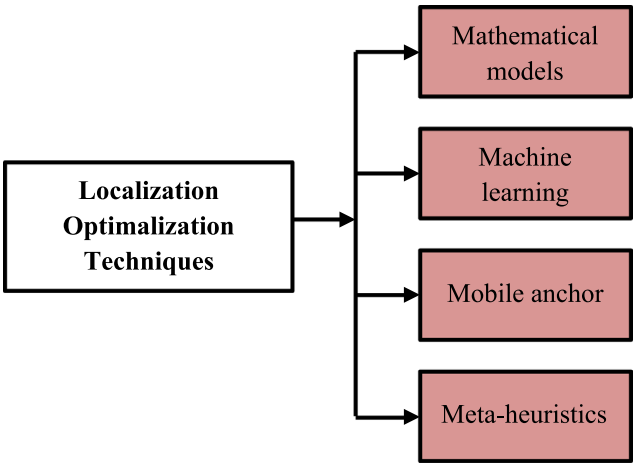


Fig. 1. Categorization of node localization optimization approaches.

2.1. Localization optimization based mathematical model

The use of mathematical models is one of the most common methods for node localization optimization in WSNs, and numerous localization strategies based on this method have been introduced in prior research [31]. These conventional approaches are categorized based on various distinguishing factors, such as network setup, mobility, computation distribution, and hardware capabilities [9]. Huang and Mao [31] research focuses on improving the accuracy and stability of the DV-Hop location algorithm. They scrutinized and rectified location errors linked to minimum hop counts by incorporating RSSI ranging technology to enhance the minimum hop phase and refines the hop distance parameters in the original algorithm. Through comprehensive simulations and analysis encompassing various beacon node ratios, node communication radii, and diverse scenarios, the study demonstrates the superiority of these enhanced algorithms over the traditional DV-Hop approach, showcasing enhanced positioning stability and accuracy. Similarly, Zhang et al. [32] addresses the issue of precise localization in WSNs with an upgraded DV-Hop algorithm that utilizes node deployment information to enhance localization precision. Simulations validate that this improved algorithm surpasses the original DV-Hop, offering improved accuracy without necessitating extra hardware. Venkata Lakshmi et al. [12] tackled the complex task of pinpointing the locations of sensor nodes within WSNs by introducing an innovative strategy that relies on a range-based localization method, casting the problem as a maximum probability distribution function. Their approach utilizes a measurement model based on received signal strength indicator and time difference of arrival, combined with the Chan algorithm, to calculate the coordinates of nodes with unknown positions. They harness ultra-wideband technology to create precise localization algorithms tailored for WSNs. To evaluate the effectiveness of the proposed system, they conducted simulations and compared the error optimization accuracy of the proposed algorithms with traditional methods found in existing literature. Their results consistently show that the proposed techniques offer superior localization accuracy when contrasted with conventional approaches. This leads to a substantial reduction in localization error, bringing it down to approximately 2.7 cm, a significant improvement over other pre-existing design. Nevertheless, the computational complexity associated with time difference of arrival renders it costly [33].

In order to reduce the computational complexity related to time difference of arrival, Lee et al. [33] employ a particle swarm optimization technique utilizing the RSSI channel model for the purpose of indoor target localization and tracking. Wang et al. [34] devised a methodology for estimating average distance that is based on probabilities and employs the probability distribution of node distances observed by an anchor node. An improvement to the DV-Hop localization algorithm called optimum anchor nodes subsets (OANS DV-Hop) was proposed by Cao and Xu [15]. They utilized optimal subset of anchor nodes and applies binary particle swarm optimization (BPSO) and continuous particle swarm optimization (PSO) to enhance localization accuracy. However, BPSO and PSO are known to be computationally intensive, especially when applied in large-scale wireless sensors networks with a significant number of nodes. A time difference of arrival technique and a Kalman filtering with DV-Hop to reduce initial errors deployment of beacons for dynamic momentum nodes was proposed for improving indoor location accuracy in WSNs [35]. In addition, the paper adopted ensemble technique particle swarm optimization with hybrid Hop algorithm to further enhance accuracy. A significant reduction in localization error was achieved compared to traditional DV-Hob with an observed error below 2% through multiple iteration. However, computational complexity and real-world scalability might be potential challenge.

Despite these techniques being widely used in this field, Alomari [36] argues that traditional approaches to node localization are becoming less feasible, compelling the need for self-localization methods. Also, Fawad et al. [37] contend that the standard range-free DV-Hop method utilizes hop distance to estimate the placements of sensor nodes, but it has limits in terms of accuracy.

2.2. Localization optimization based meta-heuristics model

In recent years, meta-heuristics have gained widespread popularity as a technique for addressing the localization challenge. These versatile optimization algorithms can be applied to a wide spectrum of problems. In the context of localization, meta-heuristics involve formulating the problem as a single or multi-objective function that is then optimized using a single or hybrid meta-heuristic algorithm. The appeal of meta-heuristics lies in their robustness and effectiveness, making them a focal point of research in WSNs localization [9,38,39]. These algorithms primarily aim to reduce position estimation errors and enhance localization accuracy. Sivakumar and Venkatesan [40] proposed different meta-heuristic methods like bat optimization algorithm with Mobile Anchor Positioning (MAP), modified cuckoo search with MAP, and firefly optimization algorithm with MAP (FOA-MAP) to address sensor node localization in wireless sensor networks. The root mean square error is used to assess performance, and experimental results demonstrate that FOA-MAP excels, minimizing localization error and outperforming all other techniques. Likewise, Alomari [36] introduces the flow direction path planning model, which utilizes real-time movement planning through meta-heuristic physics-based algorithms, surpassing conventional methods. This model delivers enhanced localization accuracy, precision, ratios, and reduced computational times compared with literature. While these studies have achieved success in this field, a potential limitation is their reliance on meta-heuristic algorithms, which can exhibit varying effectiveness based on factors like initial conditions. In the context of optimization algorithms and meta-heuristics, ‘‘initial conditions’’ refers to the initial settings or parameters of the algorithm. These conditions, such as the initial solution, randomization, parameter settings, starting point, and problem instance, establish the algorithm’s starting point and significantly influence its performance. The effectiveness and efficiency of meta-heuristic algorithms may differ based on how these initial conditions are configured. To address this, researchers often conduct sensitivity analyses and run the algorithm with various initial conditions to assess its robustness and reliability.

2.3. Localization optimization based machine learning model

An alternative method for localization optimization in WSNs involves employing machine learning. ML algorithms utilize past experiences to construct predictive models and can be categorized into supervised learning, unsupervised learning, and reinforcement learning algorithms [41]. Numerous research endeavors have been undertaken to enhance localization accuracy through the application of ML techniques. Zheng and Dehghani [42] predicted ALE based on neural network ensembles (LNNE), a range-free approach. The performance of the proposed model was compared with two established range-free localization algorithms, Centroid and DV-Hop, as well as a single neural network-based algorithm called LSNN. The results consistently showed that the proposed model outperforms the other three algorithms in terms of localization accuracy. Singh et al. [18] research introduces a support vector regression (SVR), which includes three methods (S-SVR, Z-SVR, and R-SVR) that rely on feature standardization for ALE prediction. Findings reveal strong performance across all methods, with R-SVR standing out, achieving a notable correlation coefficient of $R = 0.82$ and a low root mean square error of 0.147 m. Ref. [43] presents a rapid-SVM localization approach that enhances efficiency by streamlining support vectors. This was accomplished by establishing a minimum spanning tree using similarity metrics and clustering support vectors based on their highest similarity within the feature space. The method substitutes each cluster of support vectors with a linear combination of ‘‘determinant factor’’ and ‘‘adjusting factor’’ dictated by their similarity. Simulation results demonstrate that the algorithm reduces localization time by roughly 48% in comparison to conventional SVM-based techniques while maintaining precision.

A clustering-based localization strategies for WSN and Wireless Multimedia Sensor Network was proposed by Wajgi and Tembhurne [44]. Han et al. [6] introduced a route optimization method for WSNs using an enhanced ant colony algorithm. The experimental results of the research demonstrate that the suggested technique exhibits excellent performance in terms of energy consumption, transmission latency, number of dead nodes, and network throughput. A novel strategy for dynamic cluster-based WSN-IoT networks was proposed [45] that uses data fusion techniques to improve network resilience while eliminating irrational clustering. The proposed system fused deep learning-based data mining approach reduces data fitting while increasing the dynamic routing and balancing burden at the WSN fusion center. The empirical results show that the proposed routing algorithms outperform state-of-the-art traditional routing systems in terms of energy consumption, average latency, signaling overhead, cumulative throughput, and route heterogeneity [45]. Likewise, Cappelli et al. [46] proposed an improved localization of visible light using neural networks and optimized fingerprinting in WSNs.

Shakila and Paramasivan [47] presented a novel model of underwater sensor network sensor node localization using a range-based localization approach and the hybrid spider monkey battle royale algorithm (hybrid SMBR). The proposed system’s efficacy was measured using a variety of measures, including localization error, delivery ratio, localization coverage, energy consumption, latency, packet loss, and throughput. Cheng et al. [48] introduced a novel fusion localization approach. The study utilizes a fusion of fuzzy C-mean and K-Medoids algorithms to calculate positions using ultra-wideband technology. Additionally, it incorporates an implicit unscented particle filter-enhanced inertial navigation system to handle navigation input. In order to overcome the accuracy limitations of indoor localization using Wi-Fi, which are caused by signal instability. Zhang et al. [49] proposed an interval random analysis method to address the uncertainty in Wi-Fi-based indoor localization. The work utilized an interval random parameter lognormal shadowing model to improve the radio map and implements adaptive Bayesian comprehensive learning particle swarm optimization to boost the accuracy of position estimate.

Based on the research, determining the ALE in wireless sensor networks is a formidable yet vital undertaking that has received considerable attention. The results of these studies are very important for making localization algorithms for wireless sensor networks that are accurate and work well. Nevertheless, many approaches for ALE estimation in WSNs rely on mathematical and meta-heuristic models, which are recognized for their potential limitations [18] and may demonstrate differing effectiveness influenced by variables like initial conditions. Others used ML to tackle localization optimization in WSNs. However, as highlighted by Zheng and Dehghani [42], a potential limitation in most prior studies are the variability in the efficacy of these methods, influenced by factors including the quality and quantity of training data, as well as the choice of the most appropriate ML algorithm. According to literature [9], of all the localization algorithms mentioned earlier, none of them can attain optimal performance across all requirements, including precise positioning and minimal energy consumption. As a result, enhancements are needed to optimize the outcomes of these algorithms.

Also, it is evident that minimal attention has been devoted to investigating the impact of various ML optimization strategies on the predictive capabilities of ensemble learners. Hence, this study centers on predicting average localization error for nodes in wireless sensor networks using an optimized ensemble model. It tackles the challenge of efficiently identifying optimal ensemble machine learning parameters using different ML optimization algorithms to enhance the accuracy of ALE prediction within a limited timeframe. Ensemble learners, recognized for their inherent ability to harness the strength of multiple models, have demonstrated their potential in achieving more accurate predictions across various fields compared to traditional mathematical and meta-heuristic models. This effectiveness stems from their capacity to capture intricate patterns and relationships within data, thereby providing valuable insights for enhanced estimations in diverse domains.

3. Methodology

In this section, we explain how the different techniques were adopted to put together the proposed optimized ensemble machine-learning model for predicting the average localization error of wireless sensor networks. Fig. 2 shows our experimental flow diagram for this study. We explained each phase in subsequent sections of this paper.

3.1. Study data

The data set for this study was downloaded from Kaggle provided by Singh et al. [18]. It had four independent variables (x) and one (y) dependent variable with a total of 107 records. These independent features included the anchor ratio, transmission range, node density, and iterations. The features provided information about WSNs and the average errors Singh et al. [18] made when trying to figure out the best position for mounting a WSN. It is worth noting that there were no missing values in the dataset used for our experiment. The Min–Max function was used to scale or change any features that needed it. We split the already-processed data set into 80% training and 20% testing based on the Pareto principle, which is known to be an effective way to split training and testing data.

3.2. Hyperparameter optimization

In machine learning, choosing the most important features (hyperparameters) of a model is as key as the quality of the input data. Several methods have been suggested in the literature, but there is no proof that one is better than the others. It is important to remember that the choice of optimization technique depends on the problem, the amount of computing power available, and the data. This makes it hard for someone new to the field of ML to make the right choice. To make this argument clearer and make it easier to choose the right optimization algorithm for predicting where nodes will be in WSNs, we used three (3) well-known but less thoroughly compared optimization algorithms to optimize the hyperparameter of a random forest algorithm for predicting ALE. We aimed at selecting the optimization algorithm that enabled our ensemble model to perform best on the testing set, considering both accuracy and computational complexity. We discuss these algorithms briefly as presented in the literature [50,51].

1. **Grid search (GS):** This approach entails defining a range of potential values for each hyperparameter, after which the algorithm will train and assess a model for every possible combination of these hyperparameter values. In this work, the Grid search was implemented using the GridSearchCV class in scikit-learn.
2. **Random search (RS):** This method involves random sampling from a distribution for each hyperparameter, and the algorithm will train and evaluate a model for each set of hyperparameter values. Because it does not examine every potential combination of parameters, random search is more computationally efficient than grid search, but it still provides decent performance. We implemented Random search using the RandomizedSearchCV class in scikit-learn.
3. **Bayesian optimization (BO):** Bayesian optimization is a way to use a probabilistic model to find the global maximum or minimum of a function. It may be used to optimize the parameters of a random forest algorithm by looking for the optimal combination of parameters that maximizes the algorithm's performance on a certain dataset. This can be done by making a probabilistic model of the function being improved (in this study, the performance of the random forest algorithm) and using it to guide the search for the best parameters. The Bayesian optimization is a useful tool for improving the performance of ML models because it can quickly try out a wide range of possible parameter combinations and find the best one in fewer iterations than standard optimization methods like grid search or random search. Bayesian optimization was implemented using the BayesSearchCV class in scikit-optimize or the Hyperopt library.

3.3. Prediction model

Different ML algorithms have been used in different fields for prediction, anomaly detection, and dimensionality reduction. One such algorithm that has gained popularity of late is the ensemble learner. Ensemble learning is a technique that uses many models to improve the performance of a single model. Ensemble techniques are used a lot in ML and have been shown to improve the accuracy and stability of predictions. In this study, the random forest (RF) was adopted; it is a type of bagging algorithm, which stands for Bootstrap Aggregating. RF is an approach to learning in which many decision trees are built, and then their predictions are combined to get a final accurate prediction [52]. RF creates a group of decision trees (DT) by giving each tree a different subset of data and characteristics. The mathematical representation of RF can be viewed as the following steps:

1. Select k samples from the training data (DS) set randomly with replacement, where k is the number of samples to use to train each tree.
2. For each of the k samples, train a decision tree using a subset of the features as the split criteria.
3. Store the prediction made by each tree in a list.
4. Take the average of all the predictions made by the trees to obtain the final prediction for the Random Forest.

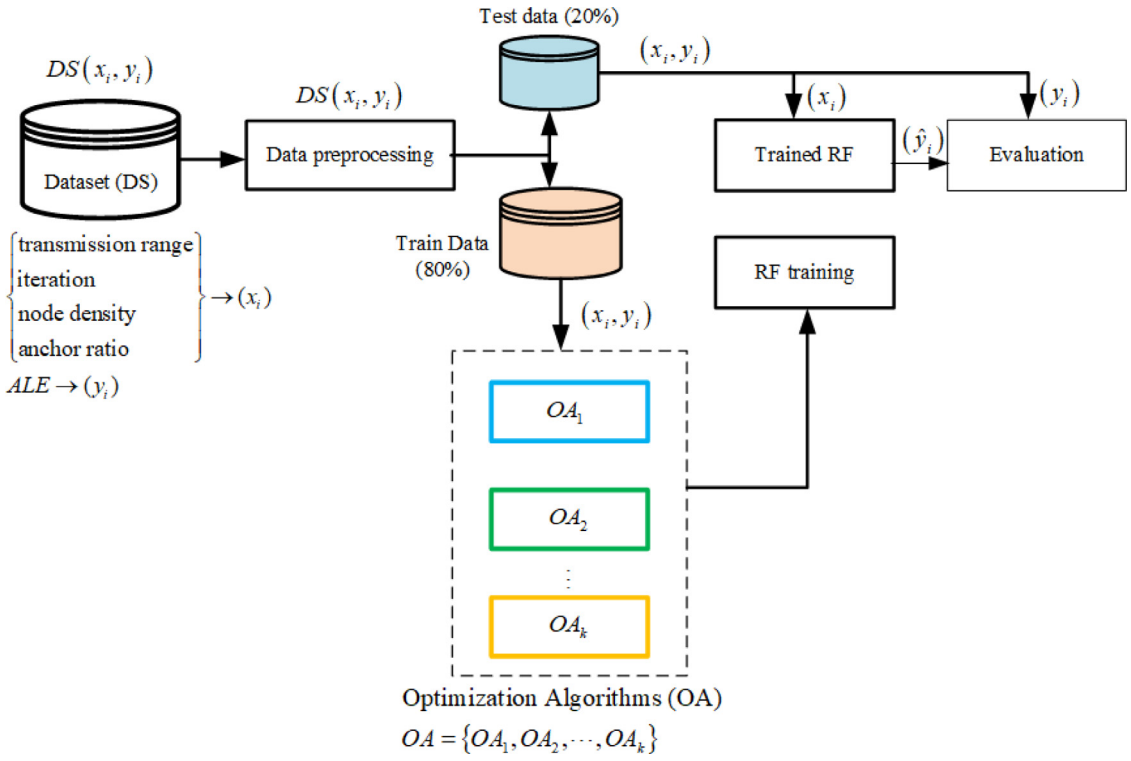


Fig. 2. Experimental flow diagram.

Mathematically, if the predictions made by each tree are represented by $y'_1, y'_2, \ldots, y'_i$, then the final prediction $\left(y'_{final}\right)$ made by the Random Forest can be represented as:

$$y'_{final} = \frac{\sum \left(y'_{1(x)}, y'_{2(x)}, \ldots, y'_{i(x)}\right)}{i} \quad (1)$$

We chose the RF because (i) it handles classification and regression problems reliably and accurately and can deal with large, high-dimensional datasets. (2) The ability to figure out how much each feature in the dataset adds to the model's accuracy, and thus how much value it has.

We used the optimized hyperparameters from each of the six (6) optimization algorithms to train the RF. The trained model is then tested on the test data, and its performance is measured and kept for comparison with the performance of the other optimization algorithms. Given a training dataset *DS* and *N* optimization algorithms (A1, A2, ...An), we optimized the hyperparameters of a random forest algorithm as shown in Algorithm 1. Fig. 3 shows the flowchart representation of algorithm 1.

3.4. Experimental setup and model evaluation

All experiments in this study were conducted using Python and its associated libraries, such as scikit-learn, pandas, hyperopt, scikit-optimize, genetic algorithm, seaborn, and matplotlib. We optimized the RF algorithm with the selected optimization algorithms discussed above using 5-fold cross-validation, and we used 'neg_mean_squared_error' as the scoring measure. We employed five (5) well-known evaluation metrics for regression ML tasks to assess the performance of our optimized RF model.

1. **Mean Absolute Error (MAE):** This metric calculates the mean deviation between expected and actual values. It is obtained by summing the absolute differences between predicted and actual values and then dividing by the sample count.
2. **Mean Squared Error (MSE):** This metric computes the mean of the squared differences between expected and actual values. It is calculated by summing the squared discrepancies between predicted and true values and dividing by the sample count.
3. **Root Mean Squared Error (RMSE):** It is the MSE's square root and a popular statistic for evaluating the performance of models.
4. **R-Squared (R^2):** It quantifies the fraction of the target variable's variation explained by the model. The R-squared value ranges from 0 to 1, where 1 indicates a perfect fit.
5. **Mean Absolute Percentage Error (MAPE):** This metric quantifies the average percentage variance between predicted and actual values. It is computed by summing the absolute disparities between predictions and actual values, dividing by actual values, and then multiplying by 100.

4. Results and discussions

The outcome of this study is divided into two sections. First, we present a brief description of the descriptive statistics of our dataset, followed by a discussion of the computational complexity of the optimization algorithms tested and the prediction from the optimized random forest.

4.1. Descriptive statistics and feature importance of study data

The table presents a comprehensive summary of the study's data, offering a detailed insight into the dataset. It provides information about key statistical aspects, including measures of central tendency, data variability, and the range of values for each variable. These statistics encompass the count (indicating the number of data points), the mean (average value), the standard deviation (indicating data dispersion), the minimum value, the 25th percentile (representing the lower quartile), the median (a measure of central tendency), the 75th percentile (representing the upper quartile), and the maximum value for each variable. Our observations reveal several key insights. The average anchor ratio stands at approximately 20.52, signifying that,

Algorithm 1

1. Initialize:
   a) Optimization algorithms $(A) = \{A_1, \cdots, A_n\}$
   b) Hyperparameters of the random forest algorithm
      i. *Number of trees (T)*
      ii. *Maximum depth (D)*
      iii. *Minimum samples per split (S)*
2. Split the training dataset (***DS)*** into a training set, X_{train} and a validation set, X_{val}, using a split ratio of *r*, such that:

$$X_{train} = r \times DS \tag{2}$$

$$X_{val} = \{(1 - r) \times DS\} \tag{3}$$

3. **For** each optimization algorithm $A_i(i = 1, 2, \cdots, n)$, perform the following steps:
   a) Use the optimization algorithm Ai to find the optimal hyperparameters (T_i, D_i, S_i) that minimize a cost function, J_i, based on *X_train*:

$$T_i, D_i, S_1 = \{\text{argmin}(J_i(\text{T}, \text{D}, \text{S}))\}$$

   where J_i is a function of the hyperparameters *T, D, S*.
   b) Evaluate the performance of the random forest algorithm on X_{val} using the optimal hyperparameters (T_i, D_i, S_i) found in step 3a.
   c) Store the results of the evaluation, $J_{val(i)}$, for each optimization algorithm Ai.
4. Select the optimization algorithm, A_i^*, that performed the best on the validation set, such that

$$A_i^* = \{argmin(J_{val(i)}) for\ i = 1, 2, \cdots N\}$$

5. Train the random forest algorithm on the entire training dataset *DS* using the optimal hyperparameters (T_i^*, D_i^*, S_i^*) found for the selected optimization algorithm A_i^*.
6. Evaluate the performance of the final trained random forest algorithm on a test set,X_{test}, which was not used during the hyperparameter optimization process, using a performance metric, *P:*

$$P = \{f(X_{test}, T_i^*, D_i^*, S_i^*)\} \tag{4}$$

where *f* is a function that measures the performance of the random forest algorithm.

on average, roughly 20.52 anchors are utilized within the dataset. The standard deviation of 6.74 underlines substantial variability around this mean. The minimum anchor ratio was 10, with the 25th percentile showing values of 15 or lower, signifying that at least a quarter of the data exhibits an anchor ratio below 15. The median anchor ratio, at 18, serves as a valuable measure of central tendency. The 75th percentile reaches 30, and the maximum anchor ratio also hits 30, indicating that most data points fall below 30, although some reach this upper limit. Moving on, the average transmission range measures approximately 17.88. With a standard deviation of 3.11, we observe a moderate degree of variation in transmission range values. The range for transmission ranges spans from a minimum of 12 to a maximum of 25, with the interquartile range (from the 25th to the 75th percentile) suggesting that a substantial portion of values clusters within this range.

The dataset also portrays an average node density of 159.81, pointing to a moderate density of nodes. However, a significant standard deviation of 71.19 highlights considerable variability in node density values. Node density ranges from a minimum of 100 to a maximum of 300, showcasing a wide spectrum of values. The interquartile range for node density extends from 100 to 200, indicating that most data points exhibit densities within this range. Additionally, the average number of iterations stands at 47.89, reflecting the typical number of iterations in the dataset. The standard deviation, measuring 24.67, indicates variability in the number of iterations. Iterations span from a minimum of 14 to a maximum of 100. The interquartile range suggests that approximately 50% of the data falls within a range of approximately 30 to 70 iterations. Finally, the average ALE registers at 0.98, suggesting that the typical ALE value hovers close to 1. The standard deviation of 0.41 points to some variability in ALE. ALE values exhibit a range from a minimum of 0.39 to a maximum of 2.57, with the interquartile range spanning approximately from 0.66 to 1.20. This indicates that most data points fall within this range (see Table 1).

Fig. 4 shows the correlation matrix among the variables in the study dataset, namely anchor_ratio, trans_range, node_density, iterations, and ALE. Notable findings include (i) ALE exhibits a negative correlation with anchor ratio (−0.074997), indicating that an elevated anchor ratio corresponds to diminished average log-likelihood estimates. (ii) ALE demonstrates a positive correlation with transmission range (0.109309), implying that larger transmission ranges are associated with higher ale values. (iii) ALE displays a robust negative correlation with node density (−0.645927), signifying that increased node density relates to lower ale values. (iv) ALE is negatively correlated with iterations (−0.400394), suggesting that greater iterations are linked to reduced ale values. These correlations underscore the substantial inverse relationship between ALE and node density, a critical insight for comprehending the dataset. Also, the correlations with anchor ratio, transmission range, and iterations offer valuable insights into the interactions between these variables and ALE.

Fig. 5 shows the significance of the different features influencing the predictive performance of the RF model. These findings underscored the respective impacts of these features on predicting the average localization error of wireless sensor networks. They provided valuable insights for optimizing these factors to enhance localization accuracy, with an awareness of the associated trade-offs. Iterations exhibited the highest importance score (0.4621), highlighting its pivotal role in predicting the average localization error. This substantial score indicated that the number of iterations was a critical factor, and variations in

Table 1
Summary statistics of study data.

	Anchor ratio	Trans range	Node density	Iterations	ALE
Mean	20.52	17.88	159.81	47.89	0.98
STD	6.74	3.11	71.19	24.67	0.41
Min	10	12	100	14	0.39
25%	15	15	100	30	0.66
50%	18	17	100	40	0.90
75%	30	20	200	70	1.20
Max	30	25	300	100	2.57

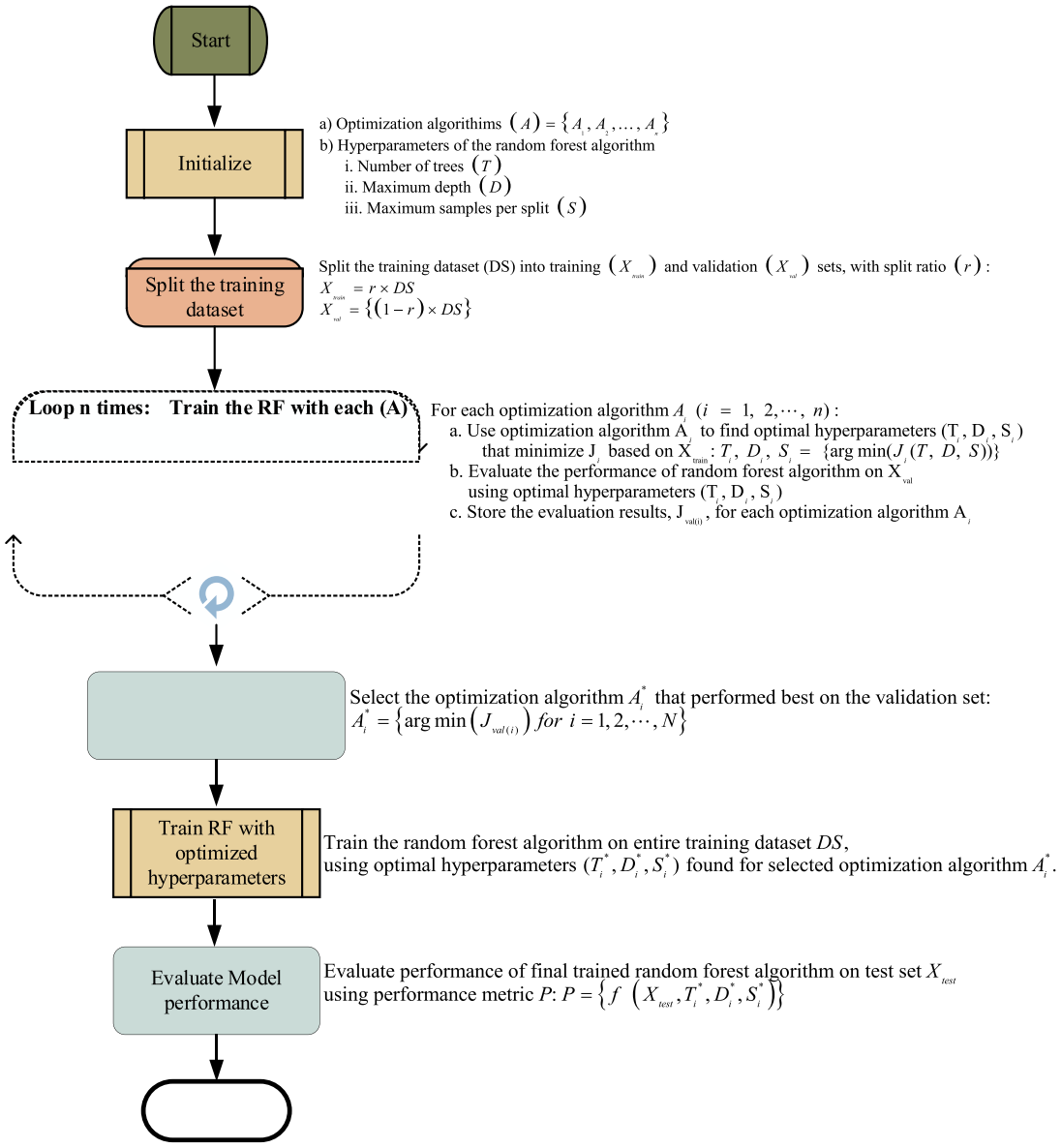


Fig. 3. Flowchart for Algorithm 1.

this feature exerted a considerable impact on the model's predictions. This feature likely represented the frequency with which the algorithm iterated to refine estimated locations, suggesting that increasing the number of iterations could lead to enhanced localization accuracy, albeit potentially at the expense of computational complexity. Additionally, "node density" got a notable importance score (0.3669), underlining the essential nature of node density within the sensor network for precise localization. Changes in node density are expected to exert a significant influence on the model's predictions. The importance of node density emphasized that the density of sensors in the network had a substantial impact on localization accuracy, implying that strategic sensor placement or node density adjustments could result in more accurate localization.

On the other hand, "anchor ratio" received a comparatively lower importance score (0.1084) than the previous features. While its influence was not as pronounced, it still contributed to the model's predictive accuracy. The anchor ratio possibly denoted the proportion of anchor nodes within the network, with variations in this feature affecting predictions. In the case of "transmission range", it garnered the lowest importance score among the features (0.0626). This indicated that changes in the transmission range had a relatively minor impact on the model's predictions in comparison to other features. Although it remained pertinent, it was not as critical as iterations or node density.

4.2. Computational complexity and best hyperparameter of compared optimization algorithms

Selecting the optimal hyperparameters is crucial for enhancing model performance. However, the decision on which optimization algorithm to employ for hyperparameter tuning should be made considering the trade-off between optimization time and quality, tailored to one's project's unique requirements. Table 2 shows the best hyperparameters determined by various optimization algorithms for fine-tuning the RF algorithm, along with their corresponding time complexities. The time complexity, denoted in seconds, represents the computational time used by each optimization algorithm. Grid Search (GS) and Random Search (RS) both took 52.74 s, indicating their swifter exploration of the hyperparameter space compared to Bayesian Optimization (BO),

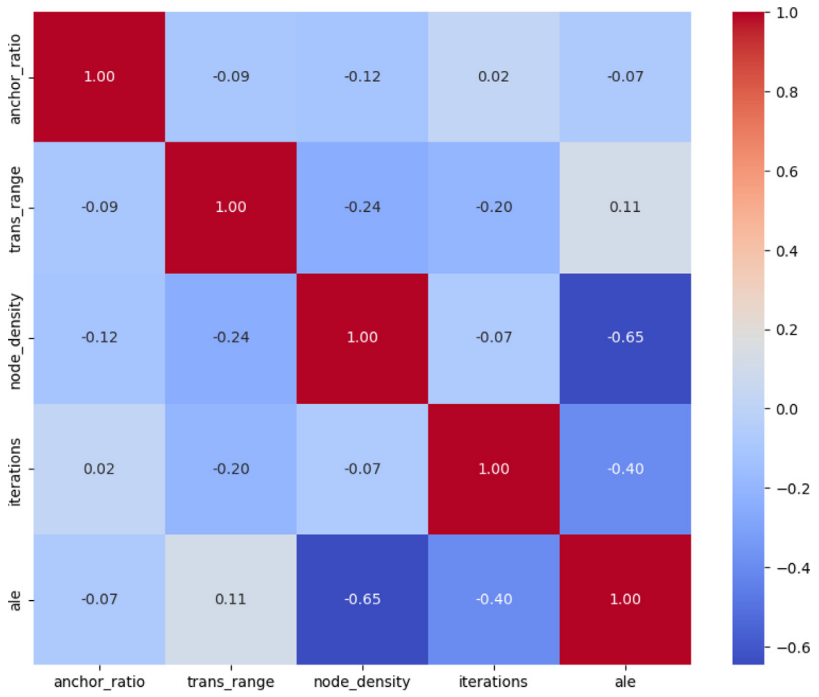


Fig. 4. Correlation matrix study data.

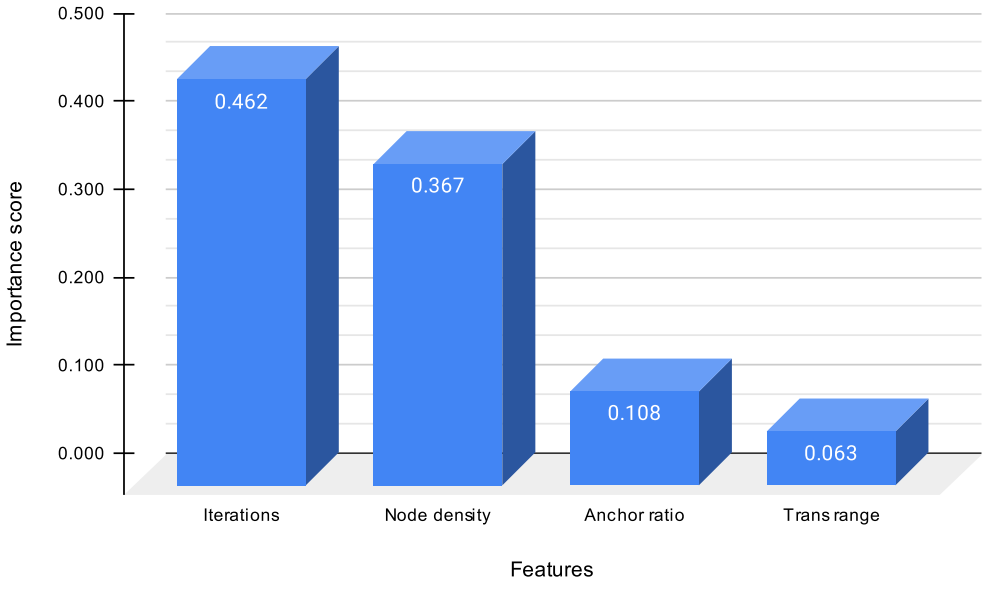


Fig. 5. Feature importance ranking.

which consumed significantly more time at 150.93 s. While Bayesian optimization took longer, it might have navigated space more effectively. Therefore, it is advisable to contemplate the trade-off between time and optimization quality, aligning it with the specific prerequisites of your project and the available computational resources.

4.3. Model prediction

Using the best hyperparameter by each optimization algorithm (see Table 2) we trained the RF algorithm to predict the average localization error of wireless sensor networks. Fig. 6 shows the performance of the Random Forest (RF) model when trained with the best hyperparameters obtained through various optimization algorithms, namely Grid Search (GS), Random Search (RS), and Bayesian Optimization (BO).

Table 2
Best hyperparameter and time complexity of the experimented optimization algorithms.

Hyperparameters	Initial range	GS	RS	BO
No._ of_estimators	(10, 400)	400	10	400
max_depth	(None, 5, 10, 15, 20)	None	10	20
min_samples_split	(2, 15)	2	2	2
min_samples_leaf	(1, 8)	1	1	2
Time complexity (seconds)	52.74	52.74	150.93	

In terms of MSE, it is evident that the random search optimized random forest (RS_RF) yielded the most favorable outcome, followed closely by grid search optimized random forest (GS_RF) with 0.046. Bayesian optimization optimized random forest (BO_RF) had a slightly

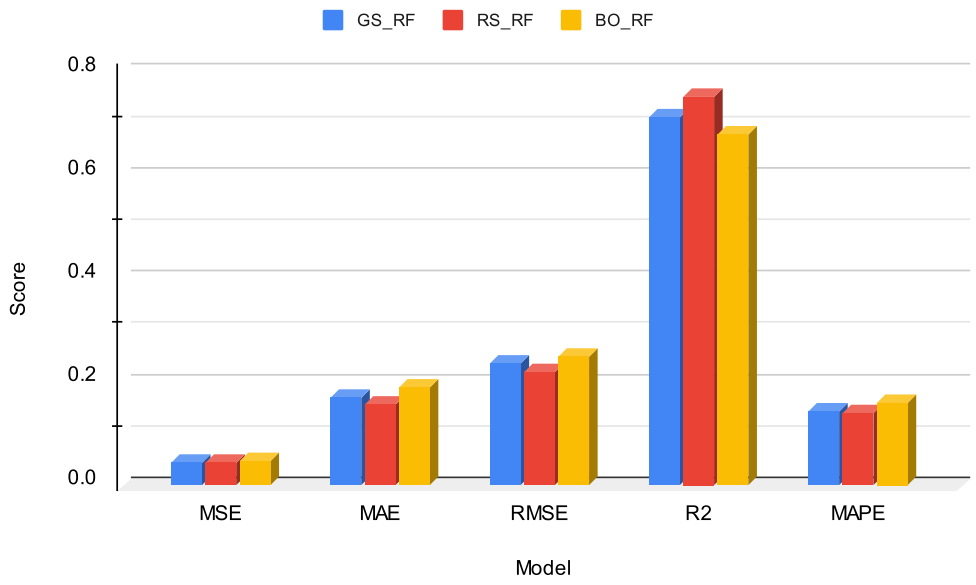


Fig. 6. Performance of the random forest model trained with the best hyperparameters.

higher MSE of 0.047. The lower MSE values directly imply greater accuracy in prediction, pointing to the enhanced predictive capabilities of RS and GS concerning the average localization error in wireless sensor networks. Similar trends emerge when considering the MAE. RS_RF outperforms the competition with an MAE of 0.156, whereas GS_RF and BO_RF achieved MAE scores of 0.171 and 0.190, respectively. This metric further underscores the accuracy of RS_RF compared to the others. The RMSE metric provides insights into prediction errors, where lower RMSE values signify superior model performance. RS_RF exhibits the lowest RMSE at 0.220, with GS_RF slightly higher at 0.238, and BO_RF trailing at 0.251. R-squared assesses how much of the variance in the dependent variable can be explained by the independent variables. RS_RF secures the highest R^2 value at 0.753, reflecting its superior data fit in contrast to GS_RF (0.712) and BO_RF (0.681). Finally, the MAPE is employed to gauge prediction accuracy as a percentage. RS_RF boasts the lowest MAPE at 0.140, followed by GS_RF at 0.145 and BO_RF at 0.159, signifying that RS_RF incurs the smallest percentage error.

The obtained results highlight the effectiveness of Random Search (RS) as the most proficient optimization algorithm for fine-tuning RF model hyperparameters in predicting the average localization error in wireless sensor networks. While Grid Search (GS) also demonstrates commendable performance across multiple metrics (MSE, MAE, RMSE, and R^2), it is outshone by RS in most aspects. Conversely, Bayesian Optimization (BO) lags slightly in performance across all metrics, indicating that, in this context, it may not be as efficient as RS and GS for hyperparameter optimization.

Table 3 presents a comparison of the performance of existing localization methods with our proposed improved random forest (RF) algorithm. Previous studies have reported root mean square error (RMSE) values ranging from 2.34 to 3.7508. However, our improved random forest (RF) model obtained a significantly lower RMSE of 0.220. Additionally, the model demonstrated a remarkable connection between anticipated and actual errors, as shown by an R2 value of 0.753. However, it is worth noting that differences in datasets between our study and those of previous works may influence direct comparisons. Our optimized RF method demonstrates superior performance compared to existing research, providing enhanced precision and dependability in localizing wireless sensor networks (WSNs). These findings have practical implications for improving decision-making in areas like environmental monitoring and smart infrastructure. They emphasize the potential of machine learning approaches in upgrading the performance of wireless sensor networks.

Table 3
A comparative analysis of proposed method with literature.

Study	Method	R^2	RMSE
Wisanmongkol et al. [53]	Ensemble deep neural network	N/A	2.34 – 3.08
Tripathy and Khilar [21]	DV-Hop and a weighted amorphous algorithm	N/A	3.0446
Shen et al. [54]	AET-amorphous	N/A	3.3211
Shen et al. [55]	Enhanced amorphous	N/A	3.7508
Proposed study	Optimized RF	0.753	0.220

5. Conclusions

This paper explored the complexities associated with estimating the ALE during the setup of wireless sensor networks (WSNs). WSNs are highly favored for real-time applications due to their compact, cost-effective, and easily deployable nature. The accurate and timely estimation of ALE is of paramount importance in WSNs, considering the potential variability both within and beyond the network, as well as the expenses related to network restructuring. Traditional approaches in the realm of WSNs have historically relied on explicit programming, which has constrained their adaptability to dynamic changes in network conditions. In response to these limitations, this research introduced a meticulously optimized ML ensemble model. The primary objectives of the study were twofold: firstly, to pinpoint the pivotal network parameters influencing node localization during the initial setup of WSNs, and secondly, to predict ALE swiftly and accurately in wireless sensor networks. The model harnessed the capabilities of the random forest algorithm, which was empowered by the fine-tuned hyperparameters obtained through diverse optimization techniques. This approach was instrumental in ALE prediction, employing essential independent variables such as node density, anchor ratio, transmission range, and iterations to bolster predictive accuracy. The findings from this research make significant contributions to the advancement of WSN technology. They introduce a more agile and effective strategy for node localization and ALE prediction, leveraging the capabilities of ML. This study opens promising avenues for WSNs to dynamically adapt to shifting network conditions. Consequently, the outcomes presented herein provide opportunities for further research and practical implementations aimed at enhancing the performance and dependability of wireless sensor networks in real-time operational scenarios.

Declaration of competing interest

We the authors declare that there is no competing interest financial or otherwise that might influence the outcome of this study.

Data availability

Data will be made available on request.

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